Lab 04 | Attack Analysis

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This case was about analyzing 5 different files: a selection of security logs, pcap (packet capture) files, and a virtual machine box (VM) that my employer believes may have been hacked while on a trip to the East Coast. I employed the use of a variety of tools for this lab, but I just began with gathering basic information about the files before proceeding further. Let us first discuss the virtual machine box.

To begin, I opened the VM box .pcap file using NetworkMiner, a built-in tool on our "win-hunt" systems. I started by obtaining basic information on the devices I was able to find on the network, one of which was particularly interesting – a Windows server. This server's name was "APLOVERS-765952", its IP address was "192.168.134.132", and it was running Windows 2000 Version 5.0 (1). There were many inbound connections to this server from other devices on the network, and it had one outbound connection itself – to another server, this time a Linux one. This Linux server's name was "1NSQhr0LMwH8Zeh3, bdl4Khd5F5KaZKSC", its IP address was "192.168.134.129", and it was running a Debian build, with a high likelihood of it being Ubuntu (2). These servers communicated with each other 6 times in the pcap's capture. Other devices on the network included a couple of hosts, which were VMs. These included "STUDENT28" and "WIN-1QO91OJGOSC", whose IP addresses were "192.168.134.1" and "192.168.134.158", respectively. When checking the DNS logs, I noticed that "WIN-1QO91OJGOSC" had requested DNS services from another server on the network, "192.168.134.2", 14 times over the course of the capture (3). Once I had this information noted, I moved on to analyzing the other files I was given.

I opened "Maybe Hacked Box—Capture.pcapng" in Wireshark and began sifting through the data for anything of note. My first idea was, again, to just check for basic information about the capture, like how long the session was, what protocols were used, etc. I found that the capture of 2394 packets went on for 4 minutes and 12 seconds on July 7th, 2017 (4). When checking for the protocols under Statistics > Protocol Hierarchy, I was able to find that IPv4, IPv6, TCP, UDP, HTTP, ARP, among others were used here. I had some information to go off now, and it was time to start filtering the packets to get a better idea of the potential situation.

The first filter I used was "http", which was just checking for HTTP connections used in the capture. I was able to confirm the connections I mentioned between the Windows and Linux servers earlier with this (5). I, then, decided to filter for UDP and TCP packets specifically to check if any other connections were made to other services. I found that by filtering for UDP packets and following the UDP stream on packets 924, 930, 1108, and 1109, Google Chrome was used to access the web (6). Also, I was able to determine what is likely our threat actor's base of operations: "192.168.134.158" or "WIN-1QO91OJGOSC". Throughout the capture, I could see this device leaving and joining multicast groups over IGMPv3. I isolated the traffic to just looking at this protocol and found that this device left and joined groups over 20 times, but the only truly "foreign" address it was communicating with was "239.255.255.250", which is the

aforementioned usage of Google Chrome (7a, 7b). I decided to keep filtering and checking through the file and stumbled upon rows upon rows of [RST, ACK] flags and alternating filter colors (8). I used the filter “frame.time_delta < 0.01” for this; this filter checks for fast connections with extremely low time between each one, and it ended up being quite important for my analysis. [RST, ACK] flags in a TCP connection mean that the port the sender was trying to access on a device was not accepting communication, and I saw a slew of ports far beyond the screenshot give this same pair of flags as a response. This led me to conclude that port scanning was occurring at the time. The source ports were also unusual and high in numerical value, easily into the 40 and 50 thousand marks, which certainly made me suspicious. All of this communication originated from the Windows server, “192.168.134.132”, which led me to suspect that the attacker was able to gain access to it in some capacity and began to use it to enumerate other ports and devices on the network.

The next filter I applied was “smb2 || smb”, which just checks for the usage of Server Message Block (SMB) protocols in the capture. SMB enables the sharing of printers, files, and ports, and in the evidence I found in this .pcapng file, the Windows and Linux servers were using SMB heavily (9). Within the SMB packets provided, I was able to determine that there were a few out-of-the-ordinary occurrences. Firstly, in packets 37 and 2260, there are two logon failures, but what was more suspicious are the packets I could see immediately afterwards (10). The info “Session Setup AndX Response, User: .\” and “Session Setup AndX Response, User: WORKGROUP\” are failed login attempts as a local user in the former, and WORKGROUP specifically in the latter. Furthermore, in the following packets, there are attempts to access the Server Service (\SRVSVC) and the NetBIOS Browser (\BROWSER) paths, which is more evidence of unauthorized access to the system (11). Not only that, but there was also an attempt to access “192.168.134.132\IPC$” on packets 40 and 2263, as well as the “\SPOOLSS” path in other packets (12). The discovery of the attempted access to “\IPC$” on the Windows 2000 server is especially significant due to the fact that \IPC$ is a share that essentially allows any user to enumerate network shares and other accounts. It is normally meant for administrative purposes in a network. The “\SPOOLSS” path is where printers and print queues are managed. Again, more evidence of enumeration.

At this point in the investigation, there was not a shadow of a doubt in my mind that this .pcapng file contained evidence of unauthorized access and lateral movement within the network. I have discovered attempts to scan ports, find devices, and access sensitive network shares that would make the previous two goals even easier to accomplish. To more convincingly cement my observations, I went to check the application log and event logs for more evidence and support that which I had already ascertained.

For the next phase of my investigation, I checked the application file for any suspicious signs and for links to the enumeration and SMB packets. Before I went any further though, I researched what event IDs would be the most obvious ones for the actions I have already seen. I started by discovering that “4625” meant there was a failed logon attempt, “4648” for any actual usage of credentials, and “5140” for any network share access, which in this case, would refer to the SMB share usage seen before. I began my search in the application log: “Maybe Hacked Box—Application Log.evt”, and found that there was no evidence of any of these on any of the logs, including the security logs. Strange. What I did find, though, is that there was evidence of the registration of “HiPerfCooker_v1” onto the server for usage of the “LocalSystem” account (13). There is also the registration of “CmdTriggerConsumer” onto the server for usage on the same account (14). Allow me to break down what is happening immediately at and around these discoveries. The “LocalSystem” account is an extensively privileged account and essentially acts as the server’s conduit to the network. Event IDs 1000 and 1001 demonstrate evidence of applications crashing on the system, which is worrying especially considering we have strange additions to the system immediately before and after these events were logged. Event ID 63 means that a “provider”, in our case performance counters, have been registered to Windows Management Instrumentation (WMI), which is the infrastructure to remotely manage and write to devices across a network. This particular context surrounding the registration of these performance counters in between them makes them quite suspicious. Also, the timestamps on these logs are dated almost a year before the .pcap files I was given, which indicates a persistent threat being established on the server long before the primary suspicious activity was detected across the network. “HiPerfCooker_V1” was registered under the “Root\WMI” path and “CmdTriggerConsumer” was registered under the “Root\cimv2” path, which further points to establishing prolonged access to the system. “CmdTriggerConsumer” is a third-party remote command execution counter going by the name “TRIGGERcmd” on the web, while “HiPerfCooker_v1” is native to Windows and does have legitimate purposes other than system monitoring. I did not believe that either of these were used legitimately, however. I decided I had found enough evidence in the application log for now and shifted gears to viewing the security logs I was provided with.

I employed the use of Event Viewer again to check through the security logs for any suspicious activity. There were 2 security logs provided to me by the CISO: “Maybe Hacked Box—Security Log.evt” and “Maybe Hacked Box—Security Log.evtx”, and I began my investigation. I went through and listed some of the event IDs I could find and began researching them to see if any of them could give me some clues on what happened. A majority of the security events I could find in both logs were dated around August 6-22nd, 2016, before a yearlong pause, which essentially confirms the presence of the attacker in the system far before the packets did any talking (15). One of the first things I noticed was that according to description for event ID 3260 , the server was added to a separate domain, “workgroup”, almost

immediately after logging began (16). This suggests that the attacker had gained access to a privileged, administrator-level account by this point already and wanted to move the server out of a potentially protected domain to be able to escalate their privileges further and move laterally in the network. This is further supported by other logs around the time this one was recorded, such as the one in screenshot (17), showing a seemingly random reboot shortly after they were able to gain access to the system in the unprotected domain (17). The event ID for this was 1074. What was also interesting about this particular log was the comment, showing that the reboot was initiated to install "VMware Tools". As I reached the end of my investigation, I wanted to use one last tool to corroborate and link the information I had to create the best possible summary and provide solid recommendations to the CISO.

Snort is a CLI-based packet analyzer, and it helped me create stronger links between pieces of information I already had. I began by just outputting the text of my general scan to a file. The command I used for the whole sequence was, "snort -c ..\etc\snort.conf -r boxcopy.pcap -A console -v > snortoutput.txt" (18). This tells Snort to use the rules in the file "snort.conf" to read the "boxcopy.pcap" file in alert (recursive) mode, be verbose, and print the output to the console and snortoutput.txt. After this, I scoured the text file for any further useful information and found a reference to the SMB message exchange between the two servers. Snort classified the rapid exchange of small TCP frames as "Potentially Bad Traffic" (19). As I mentioned before, the SMB communication between the servers is tangible evidence of the hacker escalating his privileges and moving laterally. Snort marking this exchange as suspicious traffic only strengthens this claim. This is also not a one-off flagging; there are multiple instances of this exact traffic being flagged throughout the text file of the output. Most of the output is also communication between the two servers, which is consistent with the .pcapng file I examined on Wireshark. After this point, I felt like I had enough evidence to finish my investigation and report back to my superiors on my findings and recommend some actions to take as soon as possible.

Based on my investigation into the 5 files provided, I can conclude that there was indeed an incident involving this network. The Windows 2000 server, "APLOVERS-765952", was compromised at the root level and the attacker leveraged their access to attempt logins, installations, and information access on the Linux server via SMB. This server was compromised far in advance of any attempts over the network, as the logs from the first intrusion date to August 2016, whereas the .pcaps were dated July 8th, 2017. They also at least viewed network shares, namely "IPC$", which granted them more information. They also committed the act of port scanning, which was visible in the wave of [RST, ACK] flags that can be seen in the later portions of the 4-minute pcap. The attacker was also likely able to view much of the information across the network due to their compromising of the server and registration and subsequent abuse of performance counters on it. They were able to locate many devices across the network with

their scanning and access the Internet via the Google Chrome browser on “STUDENT28”, another endpoint in the network.

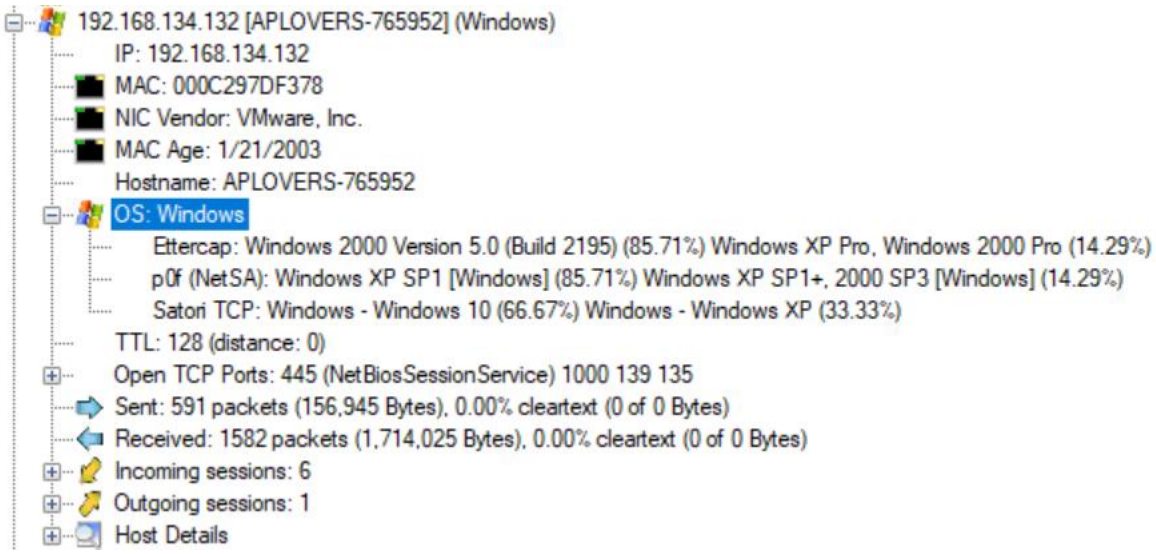
My recommendations are as follows: the security team is to be immediately briefed on the situation and sandbox these devices to isolate them from the rest of the network. I would check through the registry keys on the devices and investigate suspicious keys to prevent any further escalation or access from the attacker. They knew how to escalate privileges on the server, so I would recommend the network as a whole to be segmented, to make it more difficult for them to move laterally. I would take memory dumps and disk images of this whole network for further investigation, especially as to how they were able to get into their first base of operations and persist access to the network. This server was also running Windows 2000, so I would certainly recommend for the OS to be upgraded. The vulnerability they exploited is probably well-documented at this point, but there may not be a fix since Windows 2000 is far past its sunset date. All passwords on these devices should also be changed and the network should be monitored for any further suspicious traffic.

Tools Used

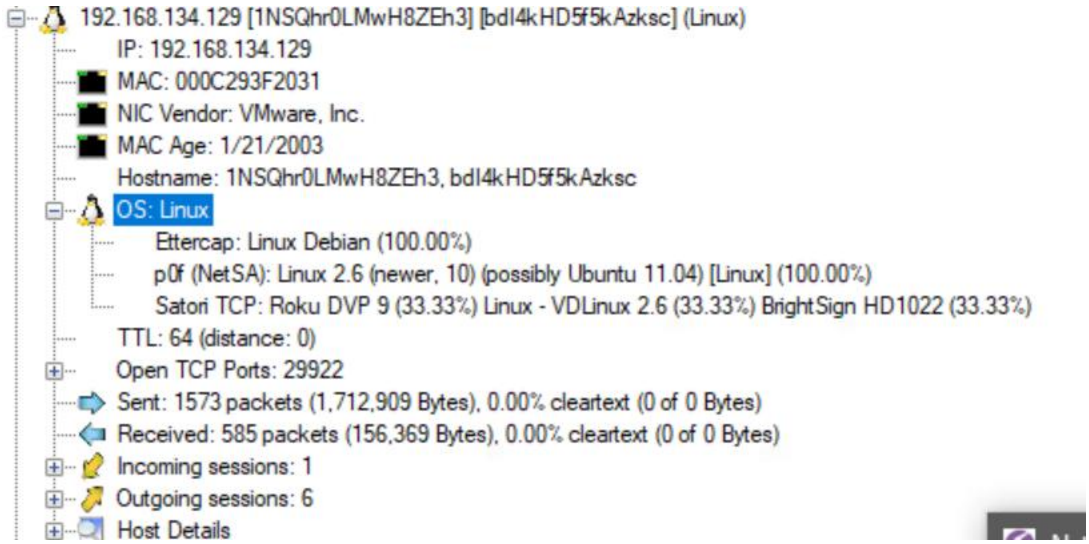
- NetworkMiner
- Wireshark
- Snort

Appendix

1.



2.



3.

Timestamp	Client	Client Port	Server	Server Port	IP TTL	DNS TTL (time)	Transaction ID	Type	DNS Query
2017-07-08 19:09:40 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	49784	192.168.134.2	53	128	00:00:05	0xEF4D	0x0001 (Host Address)	dns.msftncsi.com
2017-07-08 19:09:40 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	52069	192.168.134.2	53	128	00:00:05	0x9207	0x001C	dns.msftncsi.com
2017-07-08 19:10:42 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	49553	192.168.134.2	53	128	00:00:05	0x05CB	0x0005 (CNAME)	teredo.ipv6.microsoft.con
2017-07-08 19:10:42 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	49553	192.168.134.2	53	128	00:00:05	0x05CB	0x0005 (CNAME)	onpremwindows.ipv6.mic
2017-07-08 19:10:42 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	49553	192.168.134.2	53	128	00:00:05	0x05CB	0x0001 (Host Address)	onpremby2.ipv6.microsof
2017-07-08 19:10:45 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	58921	192.168.134.2	53	128	00:00:05	0x8C8D	0x0001 (Host Address)	dns.msftncsi.com
2017-07-08 19:10:45 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	53720	192.168.134.2	53	128	00:00:05	0xF4D6	0x001C	dns.msftncsi.com
2017-07-08 19:11:50 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	52914	192.168.134.2	53	128	00:00:05	0x8528	0x0001 (Host Address)	dns.msftncsi.com
2017-07-08 19:11:50 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	63880	192.168.134.2	53	128	00:00:05	0xA40F	0x001C	dns.msftncsi.com
2017-07-08 19:13:25 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	60781	192.168.134.2	53	128	00:00:05	0x35DC	0x0005 (CNAME)	teredo.ipv6.microsoft.con
2017-07-08 19:13:25 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	60781	192.168.134.2	53	128	00:00:05	0x35DC	0x0005 (CNAME)	onpremwindows.ipv6.mic
2017-07-08 19:13:25 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	60781	192.168.134.2	53	128	00:00:05	0x35DC	0x0001 (Host Address)	onpremby2.ipv6.microsof
2017-07-08 19:13:30 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	55315	192.168.134.2	53	128	00:00:05	0xE09C	0x0001 (Host Address)	dns.msftncsi.com
2017-07-08 19:13:30 UTC	192.168.134.158 [WIN-Q1O91OJGOSC]	58039	192.168.134.2	53	128	00:00:05	0x078E	0x001C	dns.msftncsi.com

4.

File

Name:	C:\Users\Administrator\Desktop\lab 04 stuff\Maybe Hacked Box--Capture.pcapng
Length:	2000kB
Hash (SHA256):	75461ec576473b0d00e93eda72e61d46bbcda2af613757e9324de1d216e927a3
Hash (RIPEMD160):	0cab708e2881d60b2ed231f7939578f9f9128fa5
Hash (SHA1):	5147d4dc980a1a254e38eb6fdab739fa42df2315
Format:	Wireshark/... - pcapng
Encapsulation:	Ethernet

Time

First packet:	2017-07-08 15:09:32
Last packet:	2017-07-08 15:13:45
Elapsed:	00:04:12

5.

682	8.122497	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=714429 Ack=1 Win=14600 Len=1460
683	8.122497	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=715889 Ack=1 Win=14600 Len=1460
684	8.122497	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=717349 Ack=1 Win=14600 Len=1460
685	8.122498	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=718809 Ack=1 Win=14600 Len=1460
686	8.122498	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=720269 Ack=1 Win=14600 Len=1460
687	8.122510	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=721729 Ack=1 Win=14600 Len=1460
688	8.122511	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=723189 Ack=1 Win=14600 Len=1460
689	8.122511	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=724649 Ack=1 Win=14600 Len=1460
690	8.122511	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=726109 Ack=1 Win=14600 Len=1460
691	8.122512	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=727569 Ack=1 Win=14600 Len=1460
692	8.122512	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=729029 Ack=1 Win=14600 Len=1460
693	8.122513	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=730489 Ack=1 Win=14600 Len=1460
694	8.122513	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=731949 Ack=1 Win=14600 Len=1460
695	8.122513	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=733409 Ack=1 Win=14600 Len=1460
696	8.122542	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=734869 Ack=1 Win=14600 Len=1460
697	8.122542	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=736329 Ack=1 Win=14600 Len=1460
698	8.122543	192.168.134.129	192.168.134.132	TCP	1514	29922 → 1241 [ACK] Seq=737789 Ack=1 Win=14600 Len=1460[Malformed Packet]

6.

```
HOST: 239.255.255.250:1900
MAN: "ssdp:discover"
MX: 1
ST: urn:dial-multiscreen-org:service:dial:1
USER-AGENT: Google Chrome/59.0.3071.115 Windows

M-SEARCH * HTTP/1.1
HOST: 239.255.255.250:1900
MAN: "ssdp:discover"
MX: 1
ST: urn:dial-multiscreen-org:service:dial:1
USER-AGENT: Google Chrome/59.0.3071.115 Windows

M-SEARCH * HTTP/1.1
HOST: 239.255.255.250:1900
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M-SEARCH * HTTP/1.1
HOST: 239.255.255.250:1900
MAN: "ssdp:discover"
MX: 1
ST: urn:dial-multiscreen-org:service:dial:1
USER-AGENT: Google Chrome/59.0.3071.115 Windows
```

7a.

ip.dst != 192.168.0.0/16 && ip.dst != 10.0.0.0/8 && ip.dst != 172.16.0.0/12

No.	Time	Source	Destination	Protocol	Length	Info
1783	135.343274	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
1785	135.343726	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0x8fc7 ANY WIN-Q1O91OJGOSC
1787	135.452176	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0x8fc7 ANY WIN-Q1O91OJGOSC
1793	135.764025	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
1807	143.850322	192.168.134.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
1809	144.850523	192.168.134.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
1810	145.850537	192.168.134.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
1811	146.850596	192.168.134.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
2225	233.167463	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
2227	233.171260	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
2229	233.172044	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0xf673 ANY WIN-Q1O91OJGOSC
2232	233.248266	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
2239	233.279520	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0xf673 ANY WIN-Q1O91OJGOSC
2241	233.482676	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
2243	233.484216	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
2245	233.484821	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0x26c0 ANY WIN-Q1O91OJGOSC
2248	233.591492	192.168.134.158	224.0.0.252	LLMNR	75	Standard query 0x26c0 ANY WIN-Q1O91OJGOSC

7b.

No.	Time	Source	Destination	Protocol	Length	Info
8	4.710951	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.253
9	4.711587	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
10	4.711656	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
11	4.718716	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
12	4.718839	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
13	4.721852	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
14	4.721895	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
15	4.724313	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
16	4.724435	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
21	4.755253	192.168.134.158	224.0.0.22	IGMPv3	62	Membership Report / Leave group 224.0.0.253 / Join group 224.0.0.252 for any sources
22	4.755328	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
1655	69.954048	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
1656	69.954163	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
1657	69.960825	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
1658	69.960893	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
1665	70.259084	192.168.134.158	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
1666	70.259149	fe80::9431:7662…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2

8.

frame.time_delta < 0.01

No.	Time	Source	Destination	Protocol	Length	Info
1991	230.350066	192.168.134.132	192.168.134.129	TCP	60	1723 → 52226 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1992	230.350430	192.168.134.129	192.168.134.132	TCP	74	53556 → 1755 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
1993	230.350430	192.168.134.132	192.168.134.129	TCP	60	1755 → 53556 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1994	230.350955	192.168.134.129	192.168.134.132	TCP	74	45551 → 1900 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
1995	230.350956	192.168.134.132	192.168.134.129	TCP	60	1900 → 45551 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1996	230.351359	192.168.134.129	192.168.134.132	TCP	74	52405 → 2000 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
1997	230.351360	192.168.134.132	192.168.134.129	TCP	60	2000 → 52405 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1998	230.351808	192.168.134.129	192.168.134.132	TCP	74	41726 → 1720 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
1999	230.351808	192.168.134.132	192.168.134.129	TCP	60	1720 → 41726 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2000	230.352110	192.168.134.129	192.168.134.132	TCP	74	49684 → 1582 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
2001	230.352137	192.168.134.132	192.168.134.129	TCP	60	1582 → 49684 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2002	230.352639	192.168.134.129	192.168.134.132	TCP	74	41829 → 1581 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199000 TSecr=0 WS=128
2003	230.352665	192.168.134.132	192.168.134.129	TCP	60	1581 → 41829 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2004	230.353237	192.168.134.129	192.168.134.132	TCP	74	48066 → 1533 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199001 TSecr=0 WS=128
2005	230.353239	192.168.134.132	192.168.134.129	TCP	60	1533 → 48066 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2006	230.354370	192.168.134.129	192.168.134.132	TCP	74	47192 → 1521 [SYN] Seq=0 Win=14600 Len=0 MSS=1460 SACK_PERM=1 TSval=1199001 TSecr=0 WS=128
2007	230.354397	192.168.134.132	192.168.134.129	TCP	60	1521 → 47192 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

9.

smb2 || smb

No.	Time	Source	Destination	Protocol	Length	Info
31	7.873039	192.168.134.129	192.168.134.132	SMB	154	Negotiate Protocol Request
32	7.873401	192.168.134.132	192.168.134.129	SMB	155	Negotiate Protocol Response
34	7.875834	192.168.134.129	192.168.134.132	SMB	247	Session Setup AndX Request, NTLMSSP_NEGOTIATE
35	7.876218	192.168.134.132	192.168.134.129	SMB	417	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
36	7.891686	192.168.134.129	192.168.134.132	SMB	547	Session Setup AndX Request, NTLMSSP_AUTH, User: .\
37	7.892483	192.168.134.132	192.168.134.129	SMB	105	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
38	7.897114	192.168.134.129	192.168.134.132	SMB	169	Session Setup AndX Request, User: .\
39	7.897239	192.168.134.132	192.168.134.129	SMB	158	Session Setup AndX Response
40	7.899495	192.168.134.129	192.168.134.132	SMB	143	Tree Connect AndX Request, Path: \\192.168.134.132\IPC$
41	7.899639	192.168.134.132	192.168.134.129	SMB	116	Tree Connect AndX Response
42	7.901581	192.168.134.129	192.168.134.132	SMB	161	NT Create AndX Request, Path: \SRVSVC
43	7.901704	192.168.134.132	192.168.134.129	SMB	105	NT Create AndX Response, FID: 0x0000, Error: STATUS_ACCESS_DENIED
44	7.904453	192.168.134.129	192.168.134.132	SMB	162	NT Create AndX Request, FID: 0x400a, Path: \BROWSER
45	7.904829	192.168.134.132	192.168.134.129	SMB	205	NT Create AndX Response, FID: 0x400a
46	7.912909	192.168.134.129	192.168.134.132	DCERPC	733	Bind: call_id: 0, Fragment: Single, 13 context items: 0e042bc0-cab3-517d-523f-7cbe71a19dd5 V5.1…
47	7.913008	192.168.134.132	192.168.134.129	SMB	117	Write AndX Response, FID: 0x400a, 600 bytes
48	7.914816	192.168.134.129	192.168.134.132	SMB	129	Read AndX Request, FID: 0x400a, 467 bytes at offset 562

10.

37	7.892483	192.168.134.132	192.168.134.129	SMB	105	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
38	7.897114	192.168.134.129	192.168.134.132	SMB	169	Session Setup AndX Request, User: .\

11.

2260	233.867954	192.168.134.132	192.168.134.129	SMB	105	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
2261	233.869102	192.168.134.129	192.168.134.132	SMB	177	Session Setup AndX Request, User: WORKGROUP\
2262	233.869210	192.168.134.132	192.168.134.129	SMB	158	Session Setup AndX Response
2263	233.870758	192.168.134.129	192.168.134.132	SMB	143	Tree Connect AndX Request, Path: \\192.168.134.132\IPC$
2264	233.870905	192.168.134.132	192.168.134.129	SMB	116	Tree Connect AndX Response
2265	233.872622	192.168.134.129	192.168.134.132	SMB	161	NT Create AndX Request, Path: \SRVSVC
2266	233.872680	192.168.134.132	192.168.134.129	SMB	105	NT Create AndX Response, FID: 0x0000, Error: STATUS_ACCESS_DENIED
2267	233.874531	192.168.134.129	192.168.134.132	SMB	162	NT Create AndX Request, FID: 0x400a, Path: \BROWSER

12.

35	7.876218	192.168.134.132	192.168.134.129	SMB	417	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
36	7.891686	192.168.134.129	192.168.134.132	SMB	547	Session Setup AndX Request, NTLMSSP_AUTH, User: .\
37	7.892483	192.168.134.132	192.168.134.129	SMB	105	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
38	7.897114	192.168.134.129	192.168.134.132	SMB	169	Session Setup AndX Request, User: .\
39	7.897239	192.168.134.132	192.168.134.129	SMB	158	Session Setup AndX Response
40	7.899495	192.168.134.129	192.168.134.132	SMB	143	Tree Connect AndX Request, Path: \\192.168.134.132\IPC$
41	7.899639	192.168.134.132	192.168.134.129	SMB	116	Tree Connect AndX Response
42	7.901581	192.168.134.129	192.168.134.132	SMB	161	NT Create AndX Request, Path: \SRVSVC
43	7.901704	192.168.134.132	192.168.134.129	SMB	105	NT Create AndX Response, FID: 0x0000, Error: STATUS_ACCESS_DENIED
44	7.904453	192.168.134.129	192.168.134.132	SMB	162	NT Create AndX Request, FID: 0x400a, Path: \BROWSER

13.

Level	Date and Time	Source	Event ID	Task Ca...
Information	8/6/2016 4:07:05 PM	LoadPerf	1000	None
Warning	8/6/2016 4:07:09 PM	WinMg...	63	None
Warning	8/6/2016 4:07:10 PM	WinMg...	63	None
Warning	8/6/2016 4:07:10 PM	WinMg...	63	None
Warning	8/6/2016 4:07:10 PM	WinMg...	5603	None
Warning	8/6/2016 4:07:10 PM	WinMg...	5603	None
Information	8/6/2016 4:07:12 PM	LoadPerf	1000	None

Event 63, WinMgmt

General Details

A provider, HiPerfCooker_v1, has been registered in the Windows Management Instrumentation namespace Root\WMI to use the LocalSystem account. This account is privileged and the provider may cause a security violation if it does not correctly impersonate user requests.

Log Name:	C:\Users\Administrator\Desktop\lab 04 stuff\Maybe Hacked Box--Application Log.evt		
Source:	WinMgmt	Logged:	8/6/2016 4:07:09 PM
Event ID:	63	Task Category:	None
Level:	Warning	Keywords:	Classic

14.

Level	Date and Time	Source	Event ID	Task Ca...
Warning	8/6/2016 4:08:56 PM	WinMgmt	5603	None
Information	8/6/2016 4:07:12 PM	LoadPerf	1000	None
Information	8/6/2016 4:07:12 PM	LoadPerf	1000	None
Information	8/6/2016 4:07:12 PM	LoadPerf	1000	None
Warning	8/6/2016 4:07:10 PM	WinMgmt	63	None
Warning	8/6/2016 4:07:10 PM	WinMgmt	63	None
Warning	8/6/2016 4:07:10 PM	WinMgmt	5603	None
Warning	8/6/2016 4:07:10 PM	WinMgmt	5603	None
Warning	8/6/2016 4:07:09 PM	WinMgmt	63	None
Information	8/6/2016 4:07:05 PM	LoadPerf	1001	None

Event 63, WinMgmt

General Details

A provider, CmdTriggerConsumer, has been registered in the Windows Management Instrumentation namespace Root\cimv2 to use the LocalSystem account. This account is privileged and the provider may cause a security violation if it does not correctly impersonate user requests.

15.

Information	8/6/2016 11:00:26 AM	Serial	2	None
Information	8/6/2016 11:00:32 AM	EventLog	6009	None
Information	8/6/2016 11:00:32 AM	EventLog	6005	None
Information	8/6/2016 4:05:14 PM	EventLog	6011	None
Information	8/6/2016 4:06:04 PM	Workstation	3260	None
Information	8/6/2016 4:08:17 PM	HTTP	15007	None
Information	8/6/2016 4:10:55 PM	Setup	60054	None
Information	8/6/2016 4:11:24 PM	EventLog	6005	None
Information	8/6/2016 4:11:24 PM	EventLog	6009	None
Information	8/6/2016 4:11:43 PM	SRService	115	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7035	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7036	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7036	None

16.

Level	Date and Time	Source	Event ID	Task Category
Information	8/6/2016 11:00:26 AM	Serial	2	None
Information	8/6/2016 11:00:32 AM	EventLog	6009	None
Information	8/6/2016 11:00:32 AM	EventLog	6005	None
Information	8/6/2016 4:05:14 PM	EventLog	6011	None
Information	8/6/2016 4:06:04 PM	Workstation	3260	None
Information	8/6/2016 4:08:17 PM	HTTP	15007	None
Information	8/6/2016 4:10:55 PM	Setup	60054	None
Information	8/6/2016 4:11:24 PM	EventLog	6005	None
Information	8/6/2016 4:11:24 PM	EventLog	6009	None
Information	8/6/2016 4:11:43 PM	SRService	115	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7035	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7036	None
Information	8/6/2016 4:11:52 PM	Service Control Manager	7036	None

Event 3260, Workstation

General Details

This computer has been successfully joined to domain 'workgroup'.

17.

Level	Date and Time	Source	Event ID	Task Category
Information	8/6/2016 4:13:25 PM	Service Control Manager	7035	None
Information	8/6/2016 4:13:26 PM	Service Control Manager	7035	None
Information	8/6/2016 4:13:40 PM	Service Control Manager	7036	None
Information	8/6/2016 4:13:40 PM	Service Control Manager	7035	None
Information	8/6/2016 4:13:42 PM	Tcpip	4201	None
Information	8/6/2016 4:13:55 PM	USER32	1074	None
Information	8/6/2016 4:13:55 PM	Service Control Manager	7036	None
Information	8/6/2016 4:13:56 PM	Service Control Manager	7036	None
Information	8/6/2016 4:14:02 PM	EventLog	6006	None
Information	8/6/2016 4:14:24 PM	vmci	3	None
Information	8/6/2016 4:14:32 PM	EventLog	6009	None
Information	8/6/2016 4:14:32 PM	EventLog	6005	None

Event 1074, USER32

General Details

Reason Code: 0x2
Shutdown Type: reboot
Comment: The Windows Installer initiated a system restart to complete or continue the configuration of 'VMware Tools'.

Log Name: \\ISCS-FILE\sharedfiles\Maybe Hacked Box--Security Log.evt

18.

```
C:\Users\Administrator\Desktop\Snort\bin>snort -c ..\etc\snort.conf -r Maybe Hacked Box.pcap -A console -v > snortoutput.txt
Running in IDS mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "..\etc\snort.conf"
```

19.

```
07/08-15:09:40.537398  [**] [129:12:2] Consecutive TCP small segments exceeding threshold [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP}
07/08-15:09:40.537398 192.168.134.129:41254 -> 192.168.134.132:445
TCP TTL:64 TOS:0x0 ID:49318 IpLen:20 DgmLen:147 DF
***AP*** Seq: 0x565DB66D  Ack: 0x1323FC3E  Win: 0x7B  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1143388 55468
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```